

● FLAGSHIP FIBER LASER · DUAL EXCHANGE

UltraCut Series

3 KW – 20 KW · UP TO 160 KW ON REQUEST

POWER RANGE

3 – 20 KW

The **UltraCut Series** is Wavlon's flagship platform — a fully enclosed cabin paired with dual exchangeable shuttle tables.

While one table is being cut, the operator loads the next sheet on the second table outside the cabin — slashing dead time between jobs. UltraCut spans 3 kW through 20 kW across both 3015 and 4020 frames, with power levels up to 160 kW available on request for the heaviest thick-plate fabrication.

Engineered for continuous production. Designed for the operator. Backed by Canadian sales and service.

01 — CONTINUOUS

Dual Exchange Shuttle

Load while you cut. The shuttle-table system means the machine never waits for the operator, and the operator never waits for the machine.

02 — RANGE

3 kW to 20 kW Standard

The widest power range in the Wavlon line. One platform serves thin gauge through the heaviest plate — power levels up to 160 kW available on request.

03 — FRAMES

3015 or 4020 Frame

The only Wavlon sheet series available in both frames with dual exchange. 3080 × 1540 mm or 4000 × 2000 mm work envelopes.

04 — SUPPORT

Canadian-Backed Service

Sales, installation, and lifetime support from Toronto, Ontario. CSA components and cables throughout the cabinet.

Production-line ready, from the moment it lands on your *shop floor*.



MAX LASER POWER

20 kW

Up to 160 kW on request

MAX STEEL

40 mm

Carbon steel • 20 kW

MAX LINKAGE
SPEED

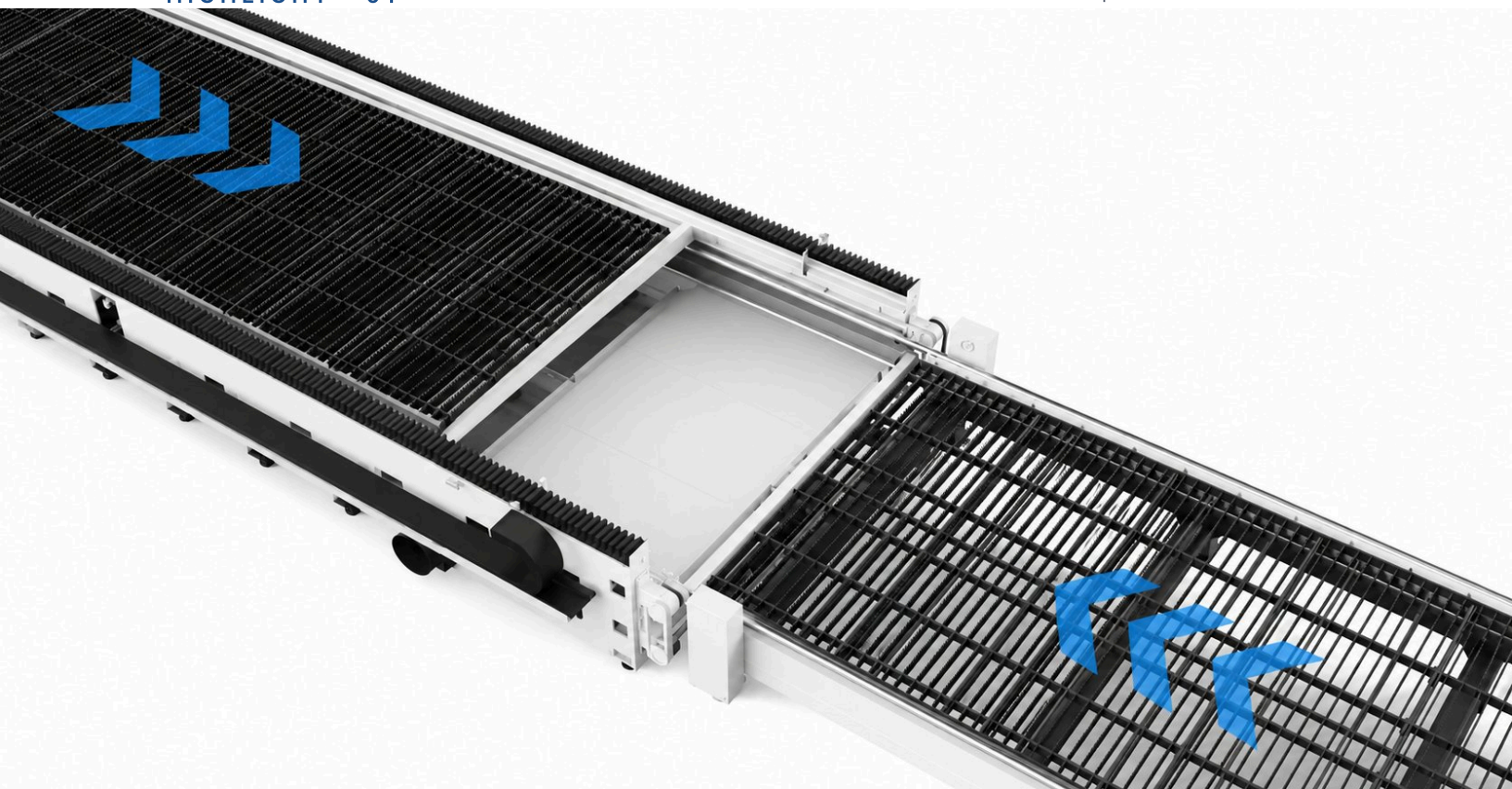
120 m/min

All power tiers

REPEAT PRECISION

±0.02 mm

Per 1000 mm



Dual Exchangeable Shuttle Tables.

Two pallets, one cycle. While the laser cuts the sheet inside the fully enclosed cabin, the operator stages the next sheet on the second table outside — completely clear of the optical path. When the cut is finished, the tables exchange automatically and the next job begins.

The result: load and unload time disappears from your cycle. The machine runs the way it was designed to — uninterrupted.

— CYCLE

01

Cut inside the cabin

Sheet A is processed under the fully enclosed cabin with active fume extraction.

02

Load outside, in parallel

Sheet B is loaded onto the second table outside the cabin while the laser is still running.

03

Exchange & resume

Tables swap pull-type. The next cut begins immediately. Operator unloads Sheet A.

Fully Enclosed Safety Cabin.

Complete optical enclosure — laser-rated viewing windows, interlocked access doors, and integrated dust and fume containment. Safe for operators, certified for the shop floor.



CABIN — AT A GLANCE

- 01 Laser-rated viewing**
Tinted safety windows let the operator monitor the cut without removing protective enclosure.
- 02 27" internal console**
Touch-screen Cypcut / Hypcut control mounted to the cabin door, operator-facing.
- 03 Blue accent lighting**
Vertical accent strips flank the cabin doors — a Wavlon design signature.
- 04 Integrated chiller cabinet**
Process water chiller and dust collector ship with the machine as a matched system.



Class-1 enclosure

Full optical containment with interlocked access doors. Safe for operation in any production environment.



27" internal console

Internal-design touchscreen with Cypcut or Hypcut control software. Full visibility of every job from the operator station.



Fume & dust extraction

Integrated extraction ports keep the cabin clear of process particulate and protect optics throughout the cut cycle.

Eyes on every cut. *Guard rails on every load.*

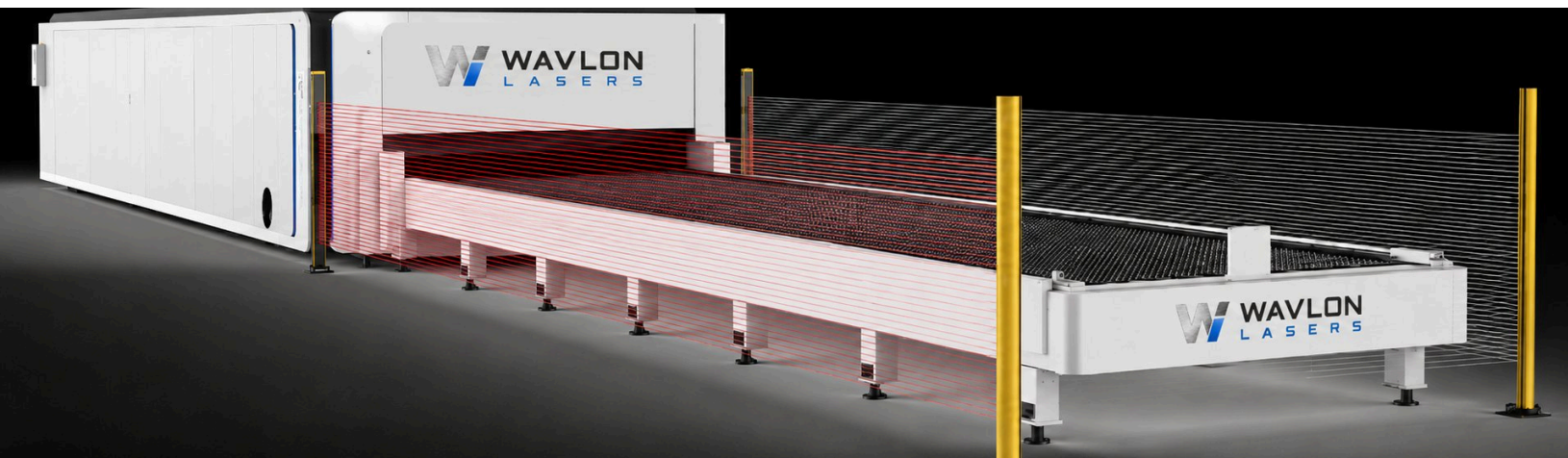
Beyond the physical cabin, UltraCut layers an electronic safety envelope around the operator — transparent inside, protected outside, without slowing the dual-exchange cycle.



01 · INSIDE THE CABIN

Intelligent Monitoring System.

UltraCut features an interconnected dual-screen system that lets operators monitor the cutting in real time. Internal high-definition cameras stream a live feed from inside the fully enclosed cabin to the operator-facing console — every cut, fully visible without breaking enclosure.



02 · OUTSIDE THE CABIN

Light Safety Curtains.

Photoelectric light curtains form an invisible safety perimeter around the shuttle-table exchange envelope. If the beam grid is broken during auto-run, motion halts instantly — operators load and unload in parallel with confidence, and the dual-exchange cycle never compromises personnel safety.

BOCI BLT Laser Heads / Precitec optional.

Four heads, one design language. Each UltraCut power tier ships standard with the matched BOCI BLT cutting head — precision adaptive standoff, automatic focus adjustment, and rapid lens exchange. **Precitec head integration is available as an option** for customers standardising on Precitec optics across their plant.

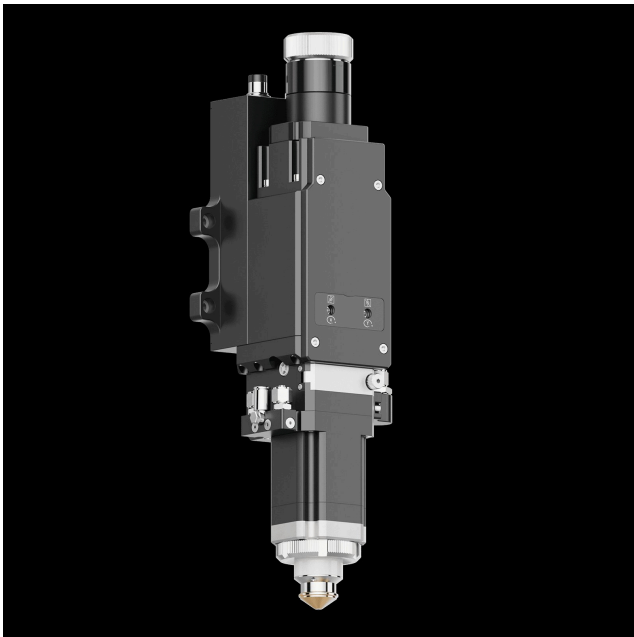


FIG. 01 – BLT CUTTING HEAD ASSEMBLY · REPRESENTATIVE MODEL

EXPLODED VIEW



PROTECTIVE LENS



POWER	BLT HEAD	NOTES
3 kW	BLT-310	Thin-gauge fidelity, fast focus
6 kW	BLT-421S	Mid-range work-horse head
12 kW	BLT-442	Thick-plate, adaptive standoff
20 kW	BLT-462	Maximum cutting range, 40 mm steel
Opt.	Precitec	Optional integration on request



Cypcut & Hypcut Control.

Industry-standard CNC software with a 27" internal-design console — nesting, edge searching, range checking, and live cut monitoring in one operator-facing interface.

- FSCUT2000E / 4000E / 8000 controllers matched to power tier
- Cypnest nesting software included standard on 12 kW and 20 kW
- Automatic calibration and one-click edge search
- 27" touchscreen — internal design, operator-facing

LASER SOURCE MAX FIBER LASER	SERVO MOTOR Yaskawa JAPAN	LINEAR GUIDE RAIL ROUST GERMANY	HELICAL RACK Eblun TAIWAN
REDUCER TECHMECH PRECISION DRIVE	ELECTRICAL Schneider · ABB FRANCE	PNEUMATIC / GAS SMC PNEUMATICS	WATER CHILLER S&A PROCESS COOLING

Choose your *power tier*.

Each UltraCut power tier is offered on both the 3015 frame (3080 × 1540 mm) and the 4020 frame (4000 × 2000 mm). All eight model codes share the same dual-exchange cabin, motion package, and component lineup.

MOST POPULAR			
ENTRY	POPULAR	MOST POPULAR	HIGH POWER
3 kW	6 kW	12 kW	20 kW
WUC-3015-3K WUC-4020-3K	WUC-3015-6K WUC-4020-6K	WUC-3015-12K WUC-4020-12K	WUC-3015-20K WUC-4020-20K
Steel 0.5–16 mm Head BLT-310 Control FSCUT2000E Nesting Optional	Steel 0.5–22 mm Head BLT-421S Control FSCUT4000E Nesting Optional	Steel 2–35 mm Head BLT-442 Control FSCUT8000 Nesting Cypnest std.	Steel 3–40 mm Head BLT-462 Control FSCUT8000 Nesting Cypnest std.

Higher power up to 160 kW — available on request.

For thick-plate fabricators requiring power beyond the standard 20 kW tier, custom UltraCut configurations are built to spec with matched cutting head, controller, and chiller package. Contact sales for configuration, lead-time, and frame options.

WUC · CUSTOM
BY QUOTATION

3015 Frame Models

Working area 3080 × 1540 mm · Standard UltraCut platform

WUC-3015

SPECIFICATION	WUC-3015-3K	WUC-3015-6K	WUC-3015-12K	WUC-3015-20K
Laser Power	3 kW	6 kW	12 kW	20 kW
Working Area	3080 × 1540	3080 × 1540	3080 × 1540	3080 × 1540
Steel Cutting Range	0.5 – 16	0.5 – 22	2 – 35	3 – 40
Laser Head	BLT-310	BLT-421S	BLT-442	BLT-462
Control System	FSCUT2000E	FSCUT4000E	FSCUT8000	FSCUT8000
X-Axis Travel	1540	1540	1540	1540
Y-Axis Travel	3080	3080	3080	3080
Z-Axis Travel	300	300	300	300
Max Acceleration	1.2 G	1.2 G	1.2 G	1.2 G
Max Linkage Speed	120 m/min	120 m/min	120 m/min	120 m/min
Nesting Software	Optional	Optional	Cypnest std.	Cypnest std.

All measurements in millimetres unless otherwise noted. **Larger and custom table sizes available on request** — the 3015 and 4020 frames are the standard offering; non-standard work envelopes can be built to spec. Contact sales@wavlonlasers.com for the most current data sheet.

4020 Frame Models

Working area 4000 × 2000 mm · Large-format UltraCut platform

WUC-4020

SPECIFICATION	WUC-4020-3K	WUC-4020-6K	WUC-4020-12K	WUC-4020-20K
Laser Power	3 kW	6 kW	12 kW	20 kW
Working Area	4000 × 2000	4000 × 2000	4000 × 2000	4000 × 2000
Steel Cutting Range	0.5 - 16	0.5 - 22	2 - 35	3 - 40
Laser Head	BLT-310	BLT-421S	BLT-442	BLT-462
Control System	FSCUT2000E	FSCUT4000E	FSCUT8000	FSCUT8000
X-Axis Travel	4050	4050	4050	4050
Y-Axis Travel	2010	2010	2010	2010
Z-Axis Travel	300	300	300	300
Max Acceleration	1.2 G	1.2 G	1.2 G	1.2 G

Common Specifications — All UltraCut Models

LASER SOURCE	MAX	MACHINE FRAME	Welded metal plate
SERVO MOTOR	Yaskawa (Japan)	MACHINE BEAM	Steel
REDUCER	TECHMECH	ENCLOSURE	Full enclosed cabin
LINEAR GUIDE RAIL	ROUST (Germany)	TABLE TYPE	Dual exchange (pull-type)
HELICAL RACK	Eblun (Taiwan)	OPERATING VOLTAGE	380 V / 3-ph / 60 Hz
ELECTRICAL	Schneider & ABB (FR)	SOFTWARE	Cypcut / Hypcut
PNEUMATIC / GAS	SMC	SCREEN	27" internal console
WATER CHILLER	S&A	CERTIFICATION	FDA & CE / CSA

Positioning precision ±0.03 mm / 1000 mm · Repeat precision ±0.02 mm / 1000 mm · Calibration automatic. **Larger and customised table sizes available on request** — contact sales for non-standard work envelopes.

ProCut • PowerCut • *UltraCut.*

Three Wavlon sheet platforms cover entry-level through flagship production. UltraCut is the only series with dual exchangeable shuttle tables — and the only platform that scales to 20 kW standard, with up to 160 kW available on request.

FLAGSHIP				
SHEET-CUTTING SERIES	WPC • ENTRY	WFC • MID	WUC • FLAGSHIP	
	ProCut	PowerCut	UltraCut	
	Entry-level job shops. Gantry beam, open bed.	Mid-tier with full safety enclosure and single auto pull-out table.	Flagship dual-exchange production. Widest power range, both frames.	
	POWER RANGE	3 – 12 kW	3 – 12 kW	3 – 20 kW • to 160 kW POA
	MAX STEEL (20 kW)	—	—	40 mm
	MAX ACCELERATION	1.0 G	1.2 G	1.2 G
	MAX LINKAGE SPEED	100 m/min	120 m/min	120 m/min
	ENCLOSURE	Gantry beam, open	Full enclosed cabin	Full enclosed cabin
	TABLE TYPE	Open single table	Single auto pull-out	Dual exchange shuttle
FRAME OPTIONS	3015 / 4020	3015 only	3015 / 4020	
BEST FOR	Entry-level job shops	Mid-tier with safety	Flagship production	

POA = Price on Application. Full spec sheets for each series are available on request from sales@wavlonlasers.com.



CANADIAN SALES & SUPPORT

Industrial-grade laser cutting, backed by Canadian *service.*

Wavlon Lasers builds, installs, and supports fiber laser cutting machines from Toronto. Sales, installation, parts, and long-term service — engineered for North American shop floors. Get a configuration and quote within one business day.

CSA COMPONENTS

1 YR ONSITE WARRANTY

PARTS WARRANTY

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